Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/18/2007
PI Name:	Javanainen, Juha D.Sc.		
Project Title:	Quantum Optics Style Theory for Degenerate Bose and Fermi Gases		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06269-3046	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007, per N. Hall-GRC (9/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2880		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In the broadest of terms, we have pursued two lines of research. First, we have developed quantum optics style theory for dilute quantum degenerate Bose and Fermi gases in the neighborhood of a Feshabch resonance. Second, we have studied the dynamics of ultralow-temperature gases in optical lattices. Our goal is to generate new theoretical concepts in the physics of degenerate atomic gases, such as a Bose-Einstein condensate or a degenerate Fermi gas realized in an alkali metal vapor.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	In the usual manner of basic theoretical research, when the concepts and ideas arising from this work circulate in the community, new theoretical ideas, experiments, and ultimately new technologies will follow. Prospective applications of interest to NASA include atomic clocks and deep-space navigation.
Task Progress:	Quite a few tasks were completed or initiated during the project period, the unifying factor being quantum-optics style theoretical methodology. Ultracold atom-molecule systems in traps: Starting from a microscopic model that includes conversion between atoms and molecules, we have developed a mean-field theory for the behavior of ultralow-temperature gas at the Feshbach resonance that occurs at certain magnetic fields. For a while the Feshbach resonance was an exceedingly fashionable field, with a surge of experiments and a flurry of theoretical effort across many areas of physics. Our theoretical methodology is by design among the simplest in this field, but in comparison with experiments it nonetheless works very well. • We have developed a time-dependent theory for fermionic atoms during the sweep of the magnetic field across a Feshbach resonance. The results were in a reasonable agreement with published experiments, and we also made several predictions that were found to agree with later experiments. • We have found the time independent solution of the time dependent theory for fermions. This turns out to be a variant of the Bardeen-Cooper-Schrieffer theory for superconductivity. An excellent agreement with immediately relevant experiments was verified. • We have developed an analogous time independent theory for bosonic atoms. No experiments exist at this time, but we predict exotic features such as a quantum phase transition when the magnetic field is varied in the neighborhood of the Feshbach resonance. • We have revisited the case of very dilute gas when the quantum mechanical effects do not yet dominate the physics . Atoms may be converted to molecules by sweeping the magnetic field across the Feshbach resonance. We have found that the atom-molecule conversion is a quantum effect; the conversion efficiency is limited by a parameter that characterizes how quantum mechanical the system is. Bosons and fermions in an optical lattice: Optical lattices, configurations of light that confine atoms and molecules into periodic structures, are essentially a novel man-made material. At the moment the main general interest is in realization of paradigmatic models of condensed-matter physics in clean and controllable experiments. • We have found a new type of instability for bosonic atoms when they flow in an optical lattice. This instability occurs when a force is applied on the atoms in an attempt to manipulate them. Counterintuitively, the smaller the force, the more prone the system is to instability. • We have earlier found that in a suitably engineered optical lattice confining fermionic atoms, the system behaves as if there were half of an atom at a defect site of the lattice. During the project period we have introduced several new experimental and engineering concepts that might facilitate future experiments on such a fractional fermion. • We have studied possible superfluidity of a combined atom-molecule gas in a rotated optical lattice. We have found the analog of anomalous flux quantization in this system. As anomalous flux quantization is a hallmark of superconductivity, we believe that the atom-molecule gas in a lattice is also a superfluid. Incidental quantum optics style topics: Based on the common underlying quantum-optics viewpoints, we have also carried out a number of small projects in a variety of areas: laser cooling and trapping, nonlinear version of the basic Landau-Zener model of quantum mechanics, development of numerical algorithms, and recent history of physics.
Bibliography Type:	Description: (Last Updated: 01/08/2020)
Articles in Peer-reviewed Journals	Javanainen J, Kostrun M, Mackie M, Carmichael A. "Simple mean-field theory for a zero-temperature fermionic gas at a Feshbach resonance." Phys Rev Lett. 2005 Sep 9;95(11):110408. PMID: 16196988 , Sep-2005
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Articles in Peer-reviewed Journals	Ishkhanyan A, Javanainen J, Nakamura H. "Landau-Zener transition in photoassociation of cold atoms: strong interaction limit." Journal of Physics A. 2006 Dec;39(48):14887-93. http://dx.doi.org/10.1088/0305-4470/39/48/004 , Dec-2006
Articles in Peer-reviewed Journals	Shevchenko A, Kaivola M, Javanainen J. "Spin-degenerate two-level atoms in on-resonance partially polarized light." Physical Review A. 2006 Mar;73(3):035801. http://dx.doi.org/10.1103/PhysRevA.73.035801 , Mar-2006
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Articles in Peer-reviewed Journals	Javanainen J. "Early days of the one-dimensional atom." Laser Physics. 2005 Oct;15(10):1428–31. , Oct-2005

Books/Book Chapters	Javanainen J. "Quantum Degenerate Gases." in "Springer Handbook of Atomic, Molecular, and Optical Physics, 2nd ed." Ed. G. Drake. New York : Springer, 2006. p. 1107-1124. http://dx.doi.org/10.1007/978-0-387-26308-3_76 , Jan-2006